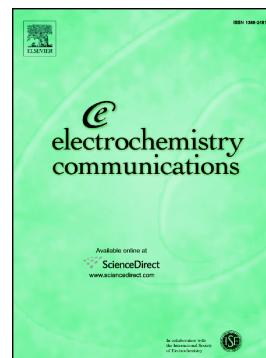


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**Strong anion receptor-assisted boron-based Mg electrolyte with wide electrochemical
window and non-nucleophilic characteristic**

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Abstract

Herein we present a strong anion receptor-assisted Mg-ion electrolyte, which is synthesized from tris(2H-hexafluoroisopropyl) borate (THFPB) and MgO in 1,2-dimethoxyethane (DME). The as-prepared borate magnesium oxide complex (BMOC) electrolyte delivers exceptional electrochemical performances, including extremely high anodic stability (up to 4.2 V vs. Mg), non-corrosivity to stainless steel and aluminium foils, and reasonable ionic conductivity of $1.74 \times 10^{-4} \text{ S cm}^{-1}$. In addition, by virtue of the non-nucleophilic characteristic of the BMOC electrolyte, S||BMOC||Mg cells have been assembled, which show a high stable discharge capacity of 1030 mAh g⁻¹ for 15 cycles and one well-defined voltage plateau of $\approx 1.1 \text{ V vs. Mg}$, yielding a desirable energy density beyond 1100 Wh kg⁻¹ based on the weight of sulfur in cathodes.

Key words:

strong anion receptor, Mg-ion electrolyte, high anodic stability, sulfur cathode, high energy density

1. Introduction

Rechargeable Mg batteries have gained more and more attention owing to their potential advantages: (1) high volumetric capacity (3833 mAh cm^{-3}); (2) high abundance in the earth crust; (3) dendrite-free Mg deposition.^[1-3] Over the past two decades, significant progresses have been made in the quest for high performance rechargeable Mg batteries, including the development of cathode materials,^[4-6] electrolytes,^[7, 8] current collectors^[9, 10] and anode materials.^[11, 12] However, the research of high performance Mg batteries still faces a major challenge: the lack of high anodically-stable and/or non-nucleophilic Mg-compatible electrolytes with non-corrosivity against non-noble current collectors.

Early studies suggest that simple-type Mg salts as commonly used in Li-ion batteries, such as $\text{Mg}(\text{ClO}_4)_2$ or $\text{Mg}(\text{BF}_4)_2$ with carbonate solvents, would react with Mg anodes to form an inert oxide film blocking the permeation of Mg^{2+} completely.^[3] The primary task is to explore passivation layer-free Mg-ion electrolyte systems, which could support reversible Mg deposition. Gregory et al. initially reported the reversible Mg deposition/dissolution process from tetrahydrofuran solutions with magnesium organoborate.^[3] Aurbach et al. later developed a large family of passivation layer-free Mg-ion electrolytes through reacting Grignard reagents with aluminum-containing Lewis acids.^[1] The electrolytes based on hexamethyldisilazide magnesium chloride (HDMSMgCl) are firstly demonstrated to be compatible with sulfur cathode.^[13] Recently, MgCl_2 and AlCl_3 mixtures have also investigated as efficient Mg-ion electrolytes.^[13-15] Unluckily these electrolytes severely corrode non-noble current collectors due to their high chloride content, resulting in a low operating voltage window less than 2.2 V vs. Mg in typical coin cells.

Searching for practical Mg-ion electrolytes with wide electrochemical window and simultaneous

compatibility with sulfur cathodes is of paramount importance. Mohtadi et al. proposed a halogen-free monocarborane Mg salt ($\text{Mg}(\text{CB}_{11}\text{H}_{12})_2/\text{tetraglyme}$), displaying high oxidative stability. However, this electrolyte was not validated to be suitable with high capacity sulfur cathode.^[16] Our group recently designed boron-centered anion-based magnesium electrolyte possessing good compatibility with sulfur cathodes.^[17] Zhao-Karger et al. also reported the conductive salt of $\text{Mg}(\text{B}(\text{HFIP})_4)_2/\text{DME}$ recently, which shed light on the future Mg-compatible electrolyte and opened a promising avenue for realizing future Mg batteries.^[18] In spite of these great progresses, new Mg-compatible electrolytes with higher anodic stability (beyond 4.0 V vs. Mg) are still highly required in order to authenticate high voltage intercalation-type cathodes. In addition, non-nucleophilic characteristic of Mg-ion electrolytes is highly desired for exploring high energy density Mg-S batteries.

Herein we firstly introduce a anion receptor to assist the formation of Mg-ion electrolyte, formed by the reaction of THFPB and MgO in DME. The BMOC electrolyte shows an extremely high anodic potential up to 4.2 V vs. Mg, highly reversible Mg deposition/dissolution, and reasonable ionic conductivity. Furthermore, high performance S||BMOC||Mg cells have been also assembled by virtue of the non-nucleophilic characteristic of the BMOC electrolyte.

2. Experimental section

The BMOC electrolytes were synthesized in Ar glove box (Mikrouna) with moisture and oxygen concentrations below 0.1 ppm at room temperature. The BMOC electrolyte was prepared by the reaction of 0.2048 g THFPB (95%, TCI (Shanghai)) and 0.0040 g MgO (99.9%, Aladdin) in 2 mL DME (anhydrous, 99.5%, Sigma), and then stirred at least 20 minutes before use. The S/C composites was prepared as our previous work.^[17] The preparation process of Chevrel Phase

Mo₆S₈ refers to previous reports.^[19] The S/C electrodes and Mo₆S₈ electrodes were prepared according to our previous work.^[17] Other characterizations including ¹H (399.65 MHz) NMR spectra, electrospray ionization mass spectrometry (ESI-MS), scanning electron microscopy (SEM), energy-dispersive spectroscopy (EDS) and X-ray diffraction (XRD) were investigated according to Ref. 17.

The electrochemical window of various current collectors was tested by linear sweep voltammetry (LSV) using electrochemical workstation (ZAHNER-Elektrik GmbH & Co. KG, Germany) at a scan rate of 5 mV s⁻¹ between open circuit potential (OCP) and 5.0 V vs. Mg. Aluminium foil, stainless steel foil, pyrolytic graphitic film, nickel foil, and copper foil serve as electrode directly. The cycling behavior of Mg deposition/dissolution was characterized via cyclic voltammograms (CV) tests. The scan rate is 5 mV s⁻¹ and the voltage range is from -1.0 to 2.0 V vs. Mg. Stainless steel serves as working electrode and Mg foil serves as both counter and reference electrodes. The ionic conductivity of the BMOC electrolytes was calculated according to our previous work.^[17] The electrochemical performances of S/C electrodes were tested by galvanostatic discharge-charge cycling at a constant current density of 10 mA g⁻¹. Cycling behavior of a symmetrical cell Mg||BMOC||Mg was conducted at different current densities of 10 μA cm⁻², 20 μA cm⁻² and 50 μA cm⁻² for 400 cycles, 10 min for charging and 10 min for discharging. The energy density were calculated according to our previous work.^[17]

3. Results and discussion

Typical CV curves (Fig. 1a) show that the BMOC electrolyte is capable of reversible Mg deposition/dissolution. In the first cycle, the cathodic peak is 1.8 mA cm⁻², and the overpotential for Mg deposition is -0.28 V. Lower over-potentials and enhanced current response are clearly

observed during the consecutive cycles, all of which illustrate the enhanced reversibility. Such phenomena can be explained by a common conditioning process extensively studied in the previous literatures.^[20, 21]

Identifying the active species is complicated owing to the chemistry evolution during the CV cycles and the failure of obtaining single-crystals in the BMOC electrolyte. It is worthy noting that the electrochemically active species in the fresh electrolyte and the cycled electrolyte is very different as demonstrated in our previously work (Ref. 17). Here, ESI-MS analyses in Fig. 2a and 2b show that the major anions and cations in the fresh electrolyte are $[(\text{DME})\text{F}_6\text{C}_2\text{HCOB}_2]^-$ and $[(\text{DME})_2\text{Mg-O-Mg}]^{2+}$, respectively. In our assumptions, MgO is probably in part dissociated by the affinity of THFPB molecules to form the non-stoichiometry DME solvated $[\text{Mg-O-Mg}]^{2+}$ cations and free oxygen radical, which would further induce the decompositions of THFPB. The probably electrolyte reactions (only including main active anions and cations) between MgO and THFPB are listed below: $\text{MgO} + \text{DME} + \text{THFPB} \rightarrow [(\text{DME})_2\text{Mg-O-Mg}]^{2+} + [(\text{DME})\text{F}_6\text{C}_2\text{HCOB}_2]^-$. Leave the fresh electrolyte overnight and a clear solution has been formed (the inset of Fig. 1a) with small amount deposits in the bottom of the electrolyte. A definitive initial concentration of the Mg^{2+} is determined to be 530 mg L^{-1} by inductively coupled plasma analysis.

In the cycled BMOC electrolyte, the efficient anion in cycled BMOC electrolyte is determined to be stable tetra(hexafluoroisopropyl)borate anion ($[\text{B}(\text{OCH}(\text{CF}_3)_2)_4]^-$) by both ESI-MS (Fig. 2c) and NMR spectra (Fig. 2d), which is in consistent with our former works.^[17] It should be noted that the mole ratio between THFPB and MgO is about 40, resulting in large amount of residual THFPB molecules after the initial THFPB-MgO reactions. Here, we assume that during the conditioning process a similar in-situ electrochemical formation mechanism might occur as that of

our previous demonstrations (Ref. 17) below: $\text{Mg} + \text{DME} + \text{THFPB} \rightarrow [(\text{DME})_n\text{Mg}]^{2+} + [\text{B}(\text{OCH}(\text{CF}_3)_2)_4]^-$, where n is in scales of 3–6.

The significance of adding MgO into the electrolyte solutions can be revealed by the fact that pure THFPB/DME solutions would undergo 8 deep CV cycles to form efficient chemistries supporting Mg stripping/plating processes, while the BMOC electrolyte shows reversible Mg stripping/plating behavior at the first CV cycle (Fig. 1b). These results demonstrate that the addition of MgO indeed plays key role in generating the incipient active species.

The oxidation onset potentials on copper, graphite film, stainless steel, and aluminum are 2.5, 3.2, 4.0, and 4.2 V vs. Mg, respectively (Fig. 1c). The anodic stability of the BMOC electrolyte supasses most Mg-compatible electrolytes.^[22-24] Fig. 1d shows that the cell could be cycled at different current densities for 400 h without a significant increase in voltage polarization. It could be observed that the temperature dependence of the ionic conductivity (Fig. 1e) is in accordance with the Arrhenius equation over the temperature range of 30–75 °C. The ionic conductivity (Fig. 1f) rises significantly with the increase of temperature, and it has a reasonable value of $1.74 \times 10^{-4} \text{ S cm}^{-1}$ at 30 °C. These above results hint that the BMOC electrolyte indeed shows promising electrochemical properties in terms of non-corrosive toward battery casings, high anodic stability and fair ionic conductivity.

Typical SEM image (Fig. 3a) of electrochemical deposits from the BMOC electrolyte shows a circular non-dendritic grains with a diameter of 2 μm . EDS result (inset in Fig. 3a,) and XRD pattern (Fig. 3b) confirms pure magnesium without impurities.

$\text{Mo}_6\text{S}_8\|\text{BMOC}\|\text{Mg}$ cells have been assembled, which displays minimal capacity fade and almost 100% coulombic efficiency for 400 cycles (Fig. 4a and 4b). The combination of Mg and sulfur can

offer a theoretical energy density of 4000 Wh L^{-1} , higher than that of a Li-S battery (2800 Wh L^{-1}).^[13, 25, 26] Fig. 4c shows the typical voltage-capacity profiles of S||BMOE||Mg battery with distinct charging platform (1.4 V vs. Mg) and discharge platform (1.1 V vs. Mg). The charge capacity for the first cycle is 520 mAh g^{-1} , but quickly increases to 920 mAh g^{-1} for the second cycle. The aforementioned electrolytic conditioning process may result in the increased capacity. Overall, the stabilized capacity is approximately 1000 mAh g^{-1} , exceeding half of the theoretical capacity (1600 mAh g^{-1} for Mg-S battery). The voltage polarization decreases with consecutive cycles. CV curve (shown in Fig. 4d) represents a highly reversible behavior and symmetric current response. Fig. 4e exhibits the published energy density in comparison with Mg-S batteries reported in literatures.^[13, 27-29] A higher capacity retention in this work compares favourably with previously reported Mg-S batteries. The above results offer strong proofs that the BMOE electrolyte has excellent compatibility with high capacity sulfur cathodes.

4. Conclusions

In summary, we proposed a novel BMOE electrolyte, which is prepared by a simple one-step dissolution route. The BMOE electrolyte presents high anodic stability (up to 4.2 V vs. Mg), high reversibility for Mg deposition/dissolution and simultaneously non-corrosivity to conventional metallic current collectors. Furthermore, good compatibility with conversion sulfur cathode indicates that the BMOE electrolytes have great prospect in rechargeable Mg-S battery systems. Further work will aim to find a suitable high potential cathode material for constructing high-energy rechargeable Mg batteries.

Acknowledgement

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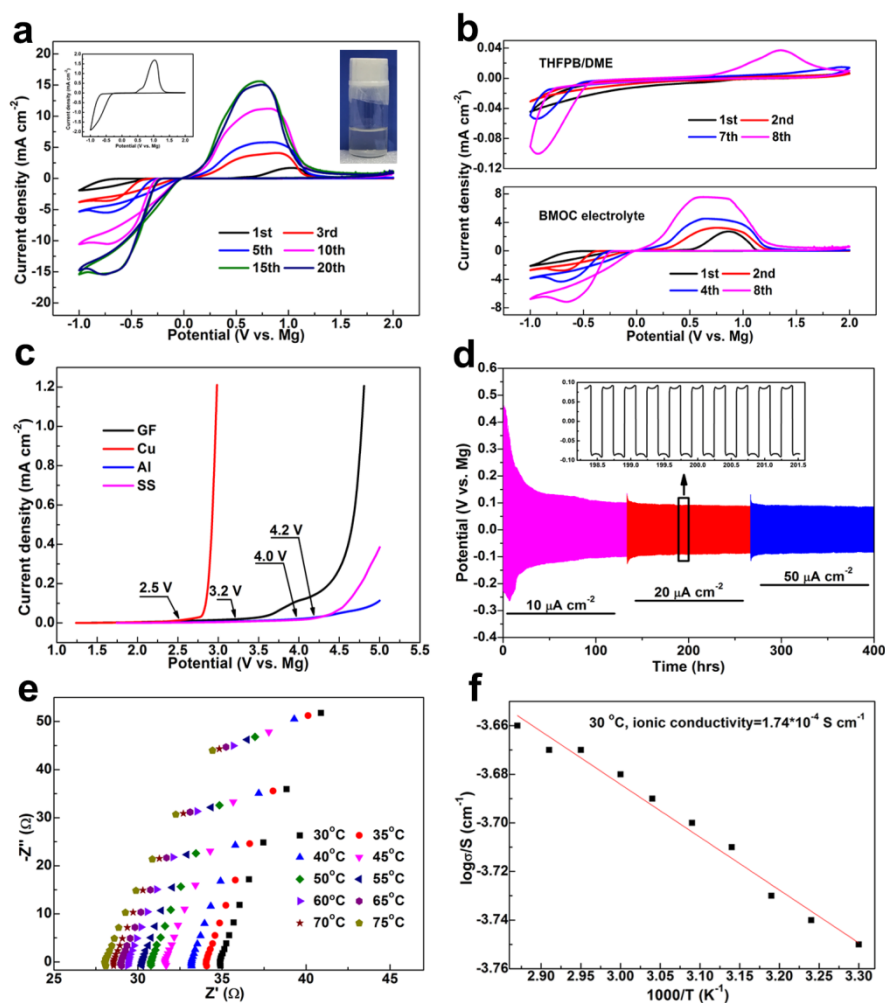


Fig. 1. (a) CV curves of Mg deposition/dissolution behaviors in BMOE electrolyte. The insets in (a) show the first CV curve and its digital photo; (b) CV curves of 0.2 mol THFPB/DME and the BMOE electrolyte; (c) LSV curves for various current collectors (GF; Cu; Al; SS); (d) Cycling behavior of a symmetrical cell at different current densities of 10 $\mu\text{A cm}^{-2}$, 20 $\mu\text{A cm}^{-2}$ and 50 $\mu\text{A cm}^{-2}$ for 400 cycles; Temperature dependence of Nyquist plots (e) and ionic conductivity profile (f) of the BMOE electrolyte.

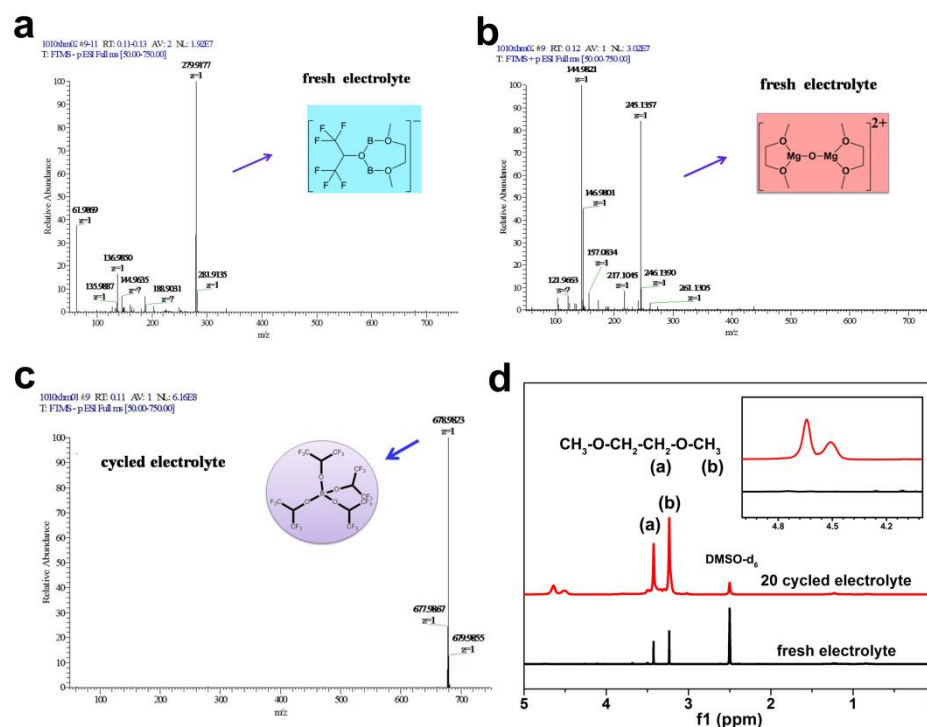


Fig. 2. MS spectra of anion species (a), cation species (b) in fresh BMOE electrolyte and anion species in cycled BMOE electrolyte (c); ^1H NMR spectra (d) of the fresh and cycled BMOE electrolytes. The inset in (d) shows the enlarged image between 5 and 4 ppm. Proton resonances at 2.51 ppm (CH₃) for DMSO, 3.25 ppm (CH₃) and 3.45 ppm (CH₂) for DME; ^1H NMR spectra appears at 4.69 ppm (CH) for $[\text{B}(\text{OCH}(\text{CF}_3)_2)_4]^-$ anions; the lower ^1H NMR spectra at 4.53 ppm can be possibly ascribed to free 1,1,1,3,3,3-hexafluoropropan-2-olate anions derived from the in-situ reaction of THFPB and Mg.

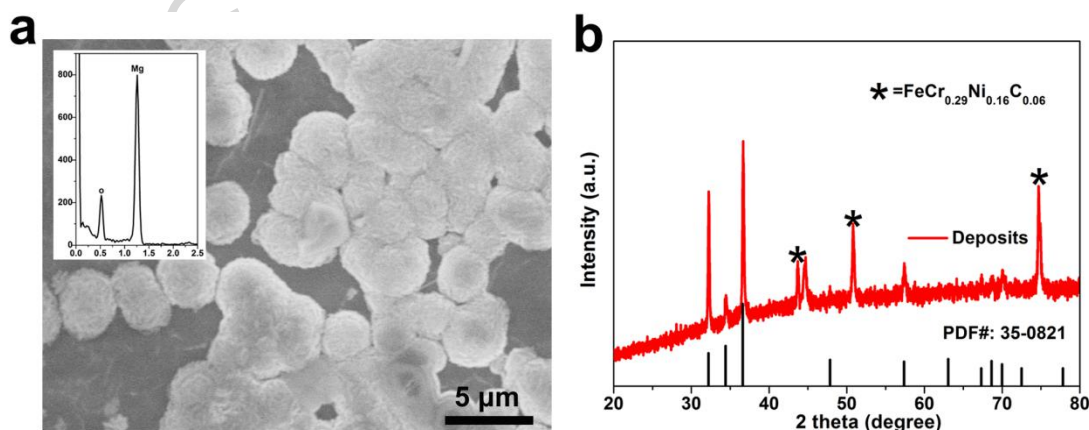


Fig. 3 Typical SEM image (a) and XRD result (b) of the deposits. The inset in (a) is the EDS

spectrum.

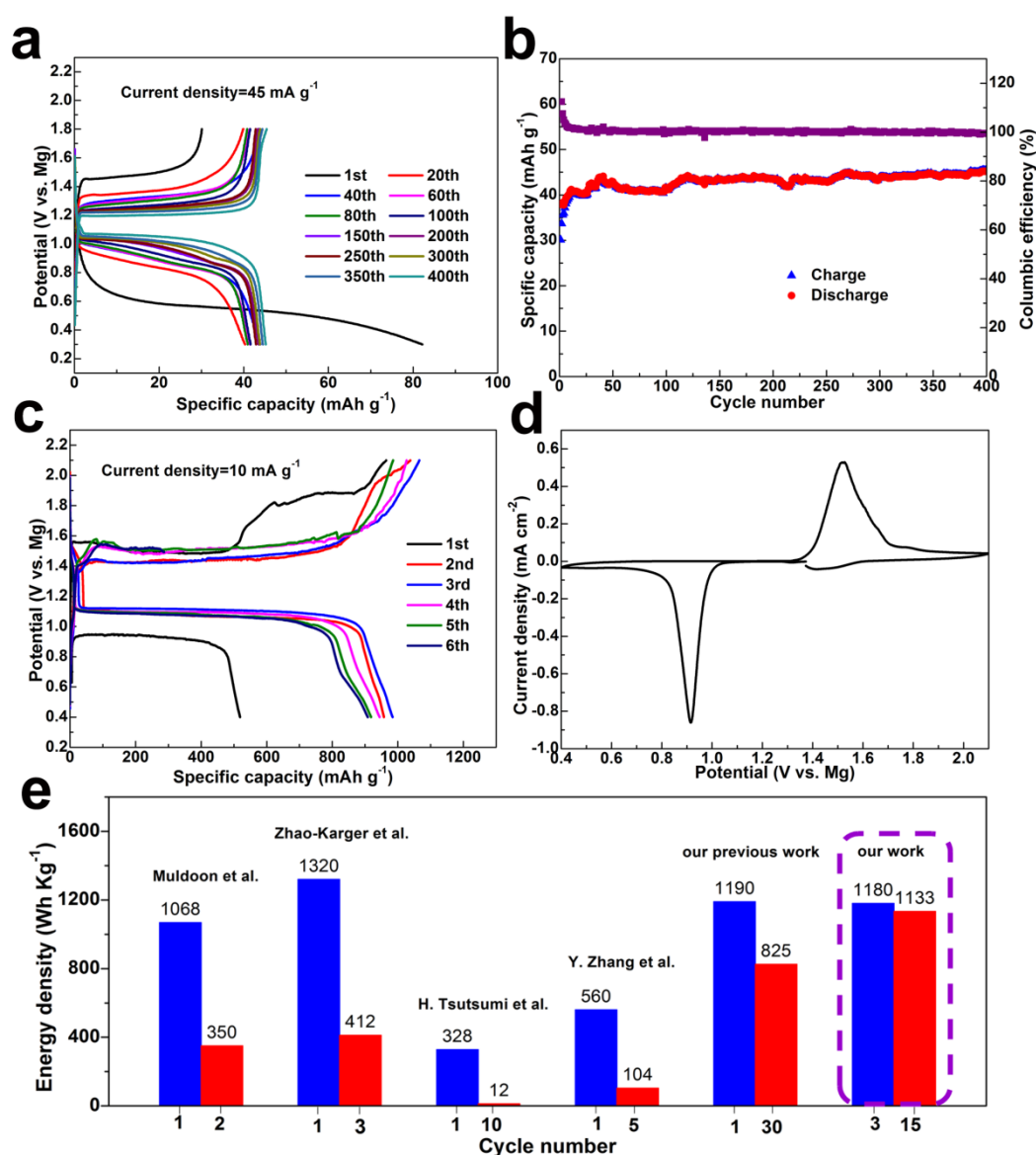
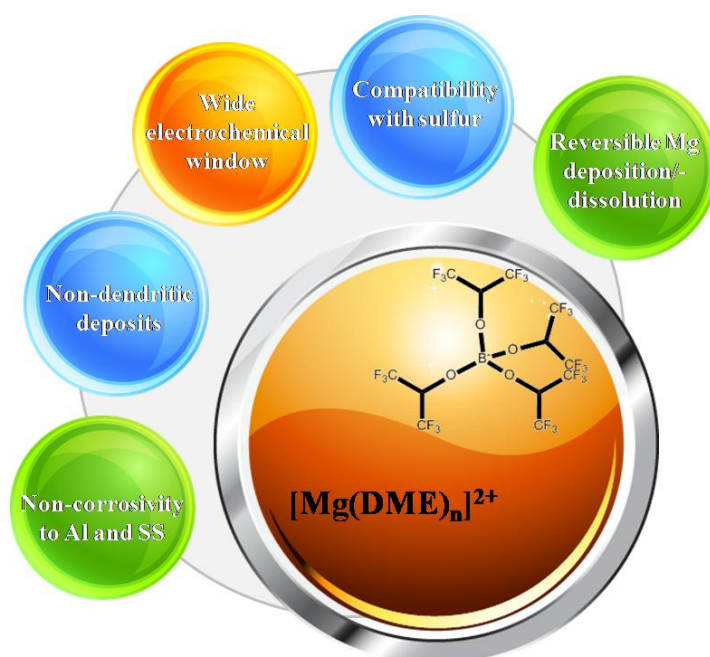


Fig. 4. Galvanostatic charge/discharge profile (a) and cycling stability (b) of Mo₆S₈||BMOC||Mg cells; Galvanostatic charge/discharge profiles (c) of S||BMOC||Mg battery; CV curves (d) of S||BMOC||Mg battery at 0.01 mV s⁻¹; (e) Comparison of energy density relating to Mg-S batteries with previous reports.

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The novel BMOC electrolyte is prepared by a simple one-step synthesis route and presents high reversibility for Mg deposition/dissolution, non-dendritic deposits, no corrosivity to conventional metallic current collectors and 4.2 V vs. Mg anodic stability on Al foil. Furthermore, good compatibility with conversion materials sulfur indicates that the BMOC electrolytes have great prospect in rechargeable Mg-S battery systems.

Highlights:

- THFPB, with powerful electron-withdrawing ability, can effectively facilitate the partial dissolution of MgO in DME forming incipient active cation/anion species.
- This newly-explored Mg-ion electrolyte delivers high anodic stability (up to 4.2 V vs. Mg).
- By virtue of the non-nucleophilic feature of the BMOC electrolyte, the as-designed Mg-S batteries display high specific capacity and relatively less attenuation.